Spectral Confounding in Watermark–Detector Interference: PSNR-Matched Controls Are Not Sufficient

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Abstract—Prior work reports that embedding a provenance watermark can harm a passive synthetic-image detector inspecting the same pixels. Testing that claim requires a control, because a detector trained on clean images responds to any imperceptible perturbation. The natural control, and the one we used initially, is a payload-free perturbation matched per image to the watermark's PSNR. On 20,000 held-out images that control appears to settle the question: matched-PSNR noise attenuates an EfficientNet-B6 detector's log-odds evidence by $\approx 16\%$ (slope 0.83–0.85) while watermarking leaves it near the clean-score identity (0.97–0.99), so the degradation is not the watermark's.

That conclusion does not survive an adequate control. PSNR fixes only the energy of a perturbation, not where that energy sits in frequency. We measure the residual spectra and find the PSNR-matched control is spectrally inverted relative to the watermark it stands in for: both tested watermarks place 0.83–0.97 of their residual power in the lowest radial band, while their PSNR-matched controls place 0.46 in the highest. We therefore build a stronger control. It is payload-free, but it carries the watermark's own per-image power spectrum through phase randomisation, and we verify it is both payload-free (recovered BER 0.51–0.52, chance) and matched (PSNR within 0.3 dB, SSIM closer to the watermark than the old control). Against it, on 800 held-out images, the gap closes on EfficientNet-B6: the watermark–control slope gap falls from $+0.126$ $[+0.105, +0.146]$ to $+0.006$ $[-0.009, +0.021]$. On this checkpoint we detect no watermark-specific interference, and the remaining gap is not distinguishable from zero at this sample size.

That closure does not generalise. We scored the identical 800-image slice through two more disclosed-provenance probes, a modern convolutional network (ConvNeXt-Base) and a vision transformer (ViT-B/16), frozen-backbone linear heads trained the same way as the ResNet-18 probe below. The spectrum-matched gap turns out to be architecture-dependent, not a property of adequate control construction in general. The modern CNN keeps a small, reliable, same-signed gap on both watermarking schemes even against the spectrum-matched control ($+0.039$ $[+0.025, +0.054]$, $+0.033$ $[+0.017, +0.052]$, both excluding zero). The transformer reverses which arm attenuates more before spectrum matching is applied at all (wm−null -0.153, -0.239, the watermark attenuating more than the PSNR-matched null, the opposite ordering from EfficientNet-B6). We report the fracture and do not have an experiment that isolates its cause.

A band-limited dose-response, run on EfficientNet-B6 only, establishes frequency-dependent detector sensitivity on that checkpoint: at fixed PSNR and zero payload, attenuation grows monotonically with the perturbation's frequency band (0.908, 0.868, 0.809 for low, mid, high). Frequency band alone does not fully account for the watermark, which attenuates less than even low-band noise. What tracks it instead is the fidelity of the spectral match: across two samples the control's slope rises from 0.838 (PSNR-only) through 0.908 (band-limited) to 0.958 (full spectrum-matched) against the watermark's 0.964, an indicative ordering rather than a single fitted trend. The original PSNR-only asymmetry is itself real, but it belongs to one checkpoint in the same narrow sense. It does not appear on the ResNet-18 probe in a full-scale, baseline-AUC-matched replication, and a resolution ablation holding architecture, data and training fixed excludes input resolution over the tested range on that pair. It does, differently, appear on the wider panel above. Two analytic results survive independently: a mean-shift statistic cannot distinguish attenuation from translation, and a strictly increasing rescaling leaves AUC exactly unchanged, so a ranking-based gate cannot register attenuation at all.

Index Terms—digital watermarking, synthetic-image detection, deepfake detection, image forensics, provenance, score calibration, AUC, checkpoint generalisation, experimental controls, pre-registration

I. Introduction

Two defences against synthetic media are converging in deployment. Proactive provenance marking embeds a signal at creation time so origin can be verified later. Passive detection infers manipulation from residual synthesis artefacts. Both read the same pixels, so they are not independent. Wu et al. [1] report that watermarking models built for genuine images can harm a deployed detector, on the argument that watermark signals overlap the forgery signals detection relies on. The same pathology is reported independently in audio [2]. If that holds generally, a detector's score distribution shifting on marked media invalidates any decision threshold calibrated on unmarked media.

That argument has a load-bearing confound. A detector trained on clean images responds to any imperceptible perturbation, so degradation after marking need not implicate the watermark's structure. No prior work in this space separates the two. We do, with a payload-free perturbation matched per image to the watermark's own PSNR, reporting every interference quantity net of that arm, with a detector-floor precondition and a gate threshold fixed before scoring.

Our pre-registered gate returned a null on ranking, and auditing why turned the result over. The watermark contributed almost nothing. The control carried the effect, and it carried it not as a shift but as an attenuation of

the detector's evidence toward zero. A strictly increasing rescaling of scores leaves AUC exactly unchanged, so a ranking gate is structurally unable to register the attenuation component of such a change. That is why a null on ranking is consistent with a real change in the score distribution.

The natural reading at that point was that watermarking is benign where imperceptible noise is not. That reading is wrong, and the control is why. PSNR constrains how much energy a perturbation carries, not where in the frequency plane it sits. Two perturbations can therefore satisfy the matching criterion while being structurally unrelated, and measuring the residual spectra shows that is what happened. Both tested watermarks put 0.83–0.97 of their residual power in the lowest radial band. The PSNR-matched controls standing in for them put 0.46 in the highest (§V-D). The control was not a weaker watermark. It was closer to the opposite perturbation with the same energy budget.

So we built the control the comparison actually needs. It is payload-free but carries the watermark's own per-image power spectrum, obtained by randomising the phase of the watermark's residual. Both properties are checked rather than assumed. The payload is unrecoverable from it (BER 0.51–0.52, chance), and it matches the watermark on PSNR to within 0.3 dB and on SSIM more closely than the control it replaces. Against it the gap closes (§V-E): the watermark–control slope gap on EfficientNet-B6 falls from $+0.126$ $[+0.105, +0.146]$, which excludes zero, to $+0.006$ $[-0.009, +0.021]$, which does not. Once the control is adequate, no watermark-specific interference is detectable at this sample size.

What remains is a real but narrower phenomenon. On EfficientNet-B6, attenuation increases monotonically with the perturbation's frequency band at fixed PSNR (§V-F), and neither watermark produces much high-frequency energy. That behaviour does not appear on a second, provenance-disclosed detector matched to it on baseline AUC (§V-C), and a resolution ablation holding architecture, training data and head fixed shows that input resolution, over the tested range, does not explain the asymmetry (§V-G). We report the mechanism as narrowed, not solved.

This paper's spine is four results. Three supporting measurements follow them and are reported as support, not as further claims.

1) The control determines the conclusion, and PSNR is not enough (§V-D). We measure what a PSNR-matched control actually matches and find it spectrally inverted relative to the watermark: 0.83–0.97 of watermark residual power in the lowest band against 0.46 in the highest for its control. This is a measurement of the standard control, not an argument about it.
2) With an adequate control, the interference effect is no longer detectable (§V-E). Against a payload-free, spectrum-matched control the watermark–control gap falls to $+0.006$ $[-0.009, +0.021]$ on EfficientNet-B6 and is not distinguishable from zero for either scheme. We verify the new arm is payload-free and matched rather than assuming it.
3) The right estimand, and a metric that cannot see it. A mean-shift statistic cannot distinguish attenuation from translation; a per-arm slope can. Separately, a strictly increasing rescaling leaves AUC exactly unchanged (unit tested), so a ranking-based gate cannot register attenuation. Both are analytic and independent of any detector.
4) The residual phenomenon is checkpoint-specific, and not explained by resolution (§V-C, §V-G). EfficientNet-B6's frequency-dependent attenuation, measured as a dose-response over three bands at fixed PSNR (§V-F), does not appear on a second detector matched to within 0.003 baseline AUC, and a resolution sweep from 224 to 448 leaves the watermark–control contrast flat at $+0.005$ to $+0.009$, against -0.126 at 528. Resolution is tested and excluded; architecture, training data and augmentation remain untested.

II. Related Work

Proactive watermarking. A substantial line of work embeds a mark at creation time to establish provenance, increasingly unifying provenance with detection or localisation [3], [4]. Our concern is orthogonal: what embedding a mark does to a passive detector inspecting the same pixels, treating watermark strength (one hand-crafted, one learned scheme) as an experimental axis rather than proposing a new one.

A larger line of work is organised around the tension we measure one instance of in §V-H. Robustness to benign processing and sensitivity to malicious editing pull in opposite directions, so a mark built to survive re-encoding is not thereby a good answer to "was this edited". Semi-fragile and separable-decoder designs target localisation explicitly rather than treating it as a by-product of robustness [5], [6], [7], [8], [9], [10], [11], [12]. A separate failure surface, diffusion-based regeneration, degrades a payload globally rather than locally [13], [14], [15]. Our splice measurement probes the local case for two specific schemes, and we do not generalise from it to either literature.

Passive detection. Reported accuracy transfers poorly once inputs are perturbed [16]; watermark robustness has been benchmarked in the same spirit [17]. Our E0 result is a sharp instance: a checkpoint advertising 98.70% accuracy scores AUC = 0.5365 on our evaluation distribution. We do not know that checkpoint's training distribution, so we report the gap rather than explain it.

Interference. The premise this work tests is Wu et al.'s [1] argument that watermarks devised for genuine images "may harm the deployed Deepfake detectors ... since the watermarks are prone to overlap with the forgery signals used for detection", answered with AdvMark, fine-tuning the watermark itself. The same pathology is reported independently in audio [2]. Guo et al. [18] benchmark

passive against watermark-based detection head-to-head but never combine them, so whether marking degrades the passive detectors being benchmarked is not asked. Two gaps follow, which this paper addresses: no prior interference result controls for generic perturbation sensitivity, and none is tested under a pre-registered null. Consequences for a fused decision rule remain open; our corpus cannot settle that (Limitations).

III. Method

A. Provenance evidence as a likelihood ratio

Let the recovered payload contain k_e bit errors out of L. Under intact provenance, errors are channel noise, $k_e \sim \mathrm{Bin}(L, p_0)$; under no valid provenance the recovered bits are uninformative, $k_e \sim \mathrm{Bin}(L, 1/2)$. The provenance log-likelihood ratio is

$$z_P = L \log 2 + k_e \log p_0 + (L - k_e) \log(1 - p_0), \quad (1)$$

with p_0 estimated on a held-out clean-transform calibration set. z_P is in log-odds units and is defined only when a mark is present.

B. Interference and the evidence-transfer estimator

For watermark scheme s and class y,

$$\Delta_\mu(s, y) = \mathbb{E}[V \mid W{=}1, S{=}s, Y{=}y] - \mathbb{E}[V \mid W{=}0, Y{=}y], \quad (2)$$

$$\Delta_{\mathrm{AUC}}(s) = \mathrm{AUC}(V; W{=}1, S{=}s) - \mathrm{AUC}(V; W{=}0). \quad (3)$$

The attenuation analyses of §V-B–V-C add one estimator: for each arm s we fit $v_s = a_s + b_s v_{\mathrm{clean}}$ by OLS, per arm and per detector separately, since the question is whether an arm's slope differs from 1, not whether arms differ from each other by assumption. Every fit is paired on the same image. Confidence intervals on b_s and on slope differences are 95% percentile bootstrap over images (resampling an image moves all its arms together), 800–2,000 resamples depending on the analysis. Where two detectors' slopes are compared (§V-C), both fits use the identical image set, so a difference in b_s cannot be an artefact of a difference in sample.

C. Fusion and abstention

$$\hat{y} = \sigma(\beta_0 + \beta_1 z_P + \beta_2 V + \beta_3 W + \beta_4 (W \cdot V)) \,. \quad (4)$$

A specification that treats the two evidence sources as independent sets $\beta_4 = 0$, so β_4 captures an interaction between provenance presence and detector evidence within this specified model. It is a model coefficient, not a direct causal measure of interference, and we read it only as such. With costs c_{FN}, c_{FP}, c_R, Chow's rule [19] gives

$$\tau_{lo} = \frac{c_R}{c_{FN}}, \qquad \tau_{hi} = 1 - \frac{c_R}{c_{FP}}, \quad (5)$$

valid iff $c_R/c_{FN} + c_R/c_{FP} < 1$, and optimal only if $\hat{y}$ is calibrated, which is the dependency this paper is about.

IV. Experimental Setup

Data. A public corpus of StyleGAN-generated faces against real photographs, balanced by class: whole-image synthesis, not face-swap or reenactment, hence "synthetic-image detection" throughout (Limitations states the cost of that substitution). The exploratory pass used 150+150 images. The confirmatory analysis scores 20,000 held-out images across five arms (clean, two schemes, two matched nulls), and §V-C adds a second detector over 8,000 of them, for 140,000 detector evaluations total, released as raw per-image scores. Splits are stratified at the image level (no identity metadata), so confidence intervals are sample-level bootstraps.

Watermarks. Two schemes spanning the hand-crafted/learned divide, pretrained: DWT-DCT-SVD (transform-domain) and RivaGAN (learned encoder/decoder), both at a 32-bit payload. Measured imperceptibility: 36.9 dB / 0.967 SSIM and 40.2 dB / 0.983 respectively. For every image and scheme we generate two payload-free controls.

The first is the standard one: additive noise binary-searched to match that image's watermark PSNR to within 0.5 dB. We call it the PSNR-matched control or the null arm.

The second is introduced here because the first is insufficient. PSNR constrains a perturbation's energy and nothing about its distribution over frequency, so an arm can satisfy it while being structurally unlike the watermark. We therefore build a spectrum-matched control: take the watermark's own residual $r = \tilde{x} - x$ on this image, compute its 2-D Fourier transform, keep the magnitude exactly and replace the phase with uniform random phase, then invert. The result carries the watermark's power spectrum per image and per channel, carries no payload, and retains no spatial structure from the mark. A scalar gain is then binary-searched to land PSNR in the same tolerance, so the two controls differ in spectrum matching and in nothing else. We verify rather than assume both properties: payload unrecoverable at chance BER, and realised PSNR/SSIM reported in §V-E.

Neither control makes the comparison perceptually or semantically exact, and we do not claim it does. What the pair licenses is a sharper statement than either alone: whether an apparent watermark effect survives as the control is matched on progressively more of the perturbation's structure.

Detectors. Six public checkpoints were screened; one, an EfficientNet-B6, clears the floor of §V-A, used zero-shot with raw logit difference $V = \ell_{\mathrm{fake}} - \ell_{\mathrm{real}}$ rather than softmax confidence. Its training data is undisclosed, so results are within-model, within-image deltas. Because one undisclosed checkpoint cannot separate a property of detection from a property of a model, we add three detectors we train ourselves: frozen, ImageNet-pretrained feature extractors with an ℓ_2-regularised logistic head fitted on the training split alone, spanning three architecture families: ResNet-18 (512-d, global-pooled), ConvNeXt-

TABLE I
Baseline AUC of every screened checkpoint on our evaluation distribution. †Model card advertises 98.70% accuracy.

Checkpoint	Baseline AUC	Status
EfficientNet-B6 (Skullly)	0.8952	clears floor
ViT (Wvolf)†	0.5365	chance
DaMsTaR/Detecto	0.5365	chance
prithivMLmods/v2	0.5299	chance
dima806	0.5296	chance
Hemg	0.4576	below chance

Base (1024-d, global-pooled) and ViT-B/16 (768-d, CLS token). Validation and test are untouched by construction for all three, and every head is deliberately linear. A weak probe over generic features differs from a purpose-trained CNN along architecture, training data and capacity at once. This is a confound we do not remove, only disclose: EfficientNet-B6 is fine-tuned on an undisclosed deepfake corpus, while the other three are frozen probes fitted only on this corpus's train split, so a difference among the four is not cleanly attributable to architecture alone, and a difference among the three probes, which share the training regime and differ only in backbone, is the cleaner comparison where one is needed. We do not claim any of the probes are the most distant possible comparison, nor that all four are architecturally comparable. Our point is only that if the effect were a property of detection rather than of one model, a panel of detectors of comparable baseline discrimination would be a place to find it, and where it appears differently across that panel (§V-E), that too is reported rather than resolved to a single verdict.

Protocol. The detector floor, the gate threshold on ΔAUC_{net}, and the fallback framing for a null were fixed before scoring. Every number below is produced by released code from a single configuration file.

V. Results

A. Detector floor check (E0)

A detector near chance makes ΔAUC a measurement of noise. We screened six public checkpoints; five of the six operate at chance on our evaluation distribution (Table I):

Two of the screened checkpoints agree to four decimal places across the same 300 images, which is consistent with one being a re-upload or a closely related checkpoint. We did not verify their provenance, so we report the coincidence rather than a conclusion about it. Among the six we screened, then, the number of independent usable detectors is smaller than the count suggests. The gap between a model card's 98.70% and AUC = 0.54 on our distribution is not a rounding error, and a floor check is what stands between such a checkpoint and a table of confident noise. This screen is a precondition for our own measurements on six checkpoints, not an audit of public detectors generally. It also bounds the study: only one public checkpoint cleared the floor, which is why we trained a second, disclosed detector rather than recording the constraint as a limitation (§V-C).

TABLE II
Confirmatory pre-registered tests, net of the control arm, with 95% percentile bootstrap intervals.

Scheme	$\Delta\mu_{net}$ (H1)	ΔAUC_{net} (H2)
DWT-DCT-SVD	$+\mathbf{0.837}$ [.807, .867]	+0.0049 [.0032, .0066]
RivaGAN	$+\mathbf{1.353}$ [1.317, 1.387]	−0.0051 [−.0071, −.0033]

TABLE III
Evidence-transfer regression, EfficientNet-B6.

Arm	slope b	intercept a
DWT-DCT-SVD	0.968	+0.125
null (PSNR-matched)	**0.834**	−0.158
RivaGAN	0.990	+0.649
null (PSNR-matched)	**0.850**	−0.125

B. Confirmatory result (n = 20,000)

Chronology, as recorded in a git-timestamped pre-registration, with each step labelled by its evidential status: (1) exploratory, a pass at n=300 on test found no effect on ranking; (2) post-hoc discovery, repairing an unrelated defect exposed $\Delta\mu_{net}$, previously unexamined, on a 36-image slice, and it was large; (3) because that observation was post-hoc and cannot be confirmed on the data that produced it, we wrote a second pre-registration committing H1/H2/H3, their thresholds and their analysis code before scoring anything further; (4) confirmatory, the pre-registered hypotheses were then tested on validation (n=20,000), which steps (1) and (2) never touched. One disclosed deviation occurred inside step (4): at 2% of the run a pipeline-verification script incidentally exposed interim values consistent with H1/H2. The run was not stopped, extended or otherwise altered in response, no threshold was changed afterwards, and the full deviation log is released with the code, so the ordering can be checked against git history rather than taken on trust.

Both pre-registered tests pass (Table II), but $\Delta\mu_{net}$ does not identify the mechanism it appears to measure. An audit shows the watermark contributes almost nothing for DWT-DCT-SVD (−0.009 watermark−clean vs. −0.846 null−clean; net +0.837 is the control falling). Nor is the effect a uniform location shift: it ranges −0.36 to +2.28 across clean-score quintiles, where a translation would be flat. Regressing each arm on the clean score separates the two components (Table III).

On this detector the null arm attenuates and the watermark arms do not. The matched-PSNR null shrinks EfficientNet-B6's log-odds evidence toward zero by $\approx 16\%$ ($b \approx 0.84$) with a small intercept ($a = -0.158/-0.125$ against a clean-score sd of 7.04), while both watermark arms keep a near-unit slope ($b = 0.968/0.990$). The watermark arms are not unchanged, however: RivaGAN carries an intercept of +0.649, i.e. a translation of the score distribution rather than a contraction of it. This is exactly the distinction the slope buys us: the null contracts, RivaGAN shifts, and $\Delta\mu$ alone would not separate them. All of this is an empirical statement about one checkpoint;

TABLE IV
Cross-detector slope comparison, 800-resample bootstrap. EffNet-B6 = EfficientNet-B6; DWT = DWT-DCT-SVD; Riva = RivaGAN. A slope below 1 indicates attenuation, so a lower b is more attenuation. Both detectors are fitted on the identical 8,000-image subset both scored, which is a subset of the 20,000 used in Table III; EfficientNet-B6 slopes therefore differ from that table in the third decimal (0.966 vs. 0.968; 0.987 vs. 0.990) because the sample differs, not the method.

Detector	Scheme	wm b [95% CI]	null b [95% CI]	null−wm
EffNet-B6	DWT	0.966 [.962, .971]	**0.832** [.825, .839]	**−.134**
EffNet-B6	Riva	0.987 [.983, .992]	**0.848** [.841, .856]	**−.139**
ResNet-18	DWT	0.956 [.951, .962]	0.971 [.967, .974]	**+.014**
ResNet-18	Riva	0.949 [.943, .954]	0.976 [.972, .979]	**+.027**

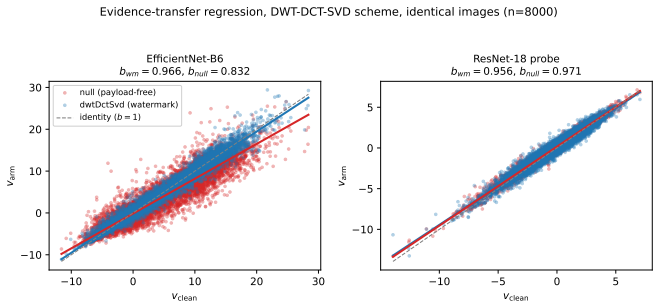


Fig. 1. Evidence-transfer regression on the same 8,000 images, DWT-DCT-SVD scheme. EfficientNet-B6 (left) shows the null arm (red) pulled below identity while the watermark (blue) tracks it; the ResNet-18 probe (right) shows the opposite ordering.

§V-C shows the ordering does not reproduce on the second detector.

Two further points are of a different kind, following from the statistics rather than from any detector. First, a strictly increasing rescaling leaves AUC exactly unchanged: for $v' = a + bv$ with $b > 0$, ranking is preserved for any a, so AUC is invariant. We verify this structural case in a unit test. It does not follow that the measured transformation is exactly affine: the regression leaves residuals, and the pre-registered ΔAUC_{net} is small but non-zero (+0.0049, −0.0051; Table II). The correct reading is that AUC cannot register the pure-attenuation component, not that an observed AUC null proves the change was pure attenuation. Second, attenuation acts on the quantity calibration depends on: shrinking log-odds toward zero changes the score-to-posterior mapping even where ranking is untouched. Whether that degrades measured calibration, and whether it moves decisions, is tested in §V-I rather than assumed here.

Effect sizes (clean-score sd 7.04): net effects of 0.12–0.19 sd, paired Cohen's d_z of 0.38/0.55, moderate, and class-dependent. The null arm erodes evidence on synthetic images roughly 3× more than on real ones.

C. Replication on a provenance-disclosed detector

One undisclosed checkpoint cannot separate "attenuation is a property of detection" from "attenuation is a property of this model". We built a detector with disclosed provenance, frozen ImageNet ResNet-18 with a logistic head fitted on training alone, and refit the regression on the same 8,000 images restricted to the identical set both detectors scored (Table IV, Fig. 1):

Read the slope as follows: $b = 1$ is no change relative to the clean score, $b < 1$ is attenuation toward zero, and a lower b is more attenuation. The ordering reverses between detectors. On EfficientNet-B6 the matched-PSNR null sits far below the watermark arm ($b = 0.832$ vs. 0.966 for DWT-DCT-SVD; 0.848 vs. 0.987 for RivaGAN): the null attenuates substantially more. On the provenance-disclosed detector the null sits slightly above the watermark arm (0.971 vs. 0.956; 0.976 vs. 0.949), so it attenuates marginally less. The intervals are tight and disjoint in both directions, so the reversal is not a resolution artefact.

The obvious objection, that a linear probe is simply too weak to express the effect, is closed by matched baseline AUC on the same images: 0.8703 (EfficientNet-B6) against 0.8725 (ResNet-18 probe), moving near-identically under every arm. The two are matched in discriminative power, not in architecture, and we claim only the former. What differs between them is how the score distribution moves, which is what the regression measures and AUC cannot see.

We therefore withdraw "imperceptible noise attenuates detector evidence while watermarking does not" as a claim about passive detection, and retain it only as a description of EfficientNet-B6. Why that checkpoint behaves this way is not settled here. Training augmentation and its 528×528 processing resolution against the probe's 224×224 are both plausible, and neither is tested. Two consequences survive. The control arm is more strongly motivated than before, not less: without it a study reports the null's −0.846 shift as the watermark's, and with it but only one checkpoint it reports a model-specific artefact as a property of detection. And the two structural results, that $\Delta\mu$ conflates attenuation with translation and that AUC is exactly blind to monotone attenuation, are analytic. They constrain what any such experiment can detect, and no replication outcome, ours included, bears on them.

D. What a PSNR-matched control actually matches

Everything above rests on the PSNR-matched control, so before drawing conclusions from it we measured what it matches. For each image we form the residual of each arm against the same clean image, take its 2-D power spectrum, and integrate over three equal-width radial bands. Fractions are per image, so arms matched on energy by construction remain distinguishable by where that energy sits.

The PSNR-matched control is spectrally inverted relative to the watermark. Both watermarks put the large majority of their residual power in the lowest band (0.825 and 0.967); the controls standing in for them put theirs in the highest (0.461, 0.455). Satisfying a PSNR constraint

TABLE V
Fraction of residual power by radial frequency band ($n = 100$), with payload BER measured on the same arms. BER near 0.5 is chance, i.e. the arm carries no recoverable payload.

Scheme	Arm	low	mid	high	BER
DWT-DCT-SVD	watermark	**0.825**	0.071	0.104	0.0022
DWT-DCT-SVD	control (PSNR)	0.143	0.396	**0.461**	0.5072
DWT-DCT-SVD	control (spectrum)	**0.824**	0.073	0.103	0.5119
RivaGAN	watermark	**0.967**	0.024	0.008	0.0013
RivaGAN	control (PSNR)	0.153	0.391	**0.455**	0.4947
RivaGAN	control (spectrum)	**0.962**	0.027	0.011	0.5194

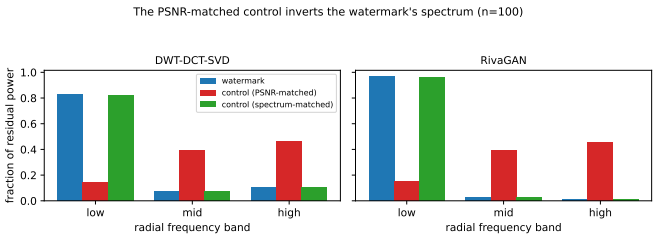


Fig. 2. The PSNR-matched control is spectrally inverted relative to the watermark it stands in for; the spectrum-matched control is not.

left the perturbation free to occupy the opposite end of the spectrum, and it did. The spectrum-matched control reproduces the watermark's profile to within 0.005 in every band while remaining at chance BER.

This is a property of the matching criterion, not of our noise model in particular. Any construction that fixes only energy admits the same freedom.

E. With an adequate control, the effect fractures along architecture

We re-ran the interference measurement with both controls present, on 800 held-out test images at an index offset disjoint from every earlier analysis, scoring seven arms per image. EfficientNet-B6 and the ResNet-18 probe were scored together. ConvNeXt-Base and ViT-B/16 were scored in a separate pass over the identical image indices and merged after verifying the two passes covered the same 800 images with no overlapping detector names (merge_e7_results.py). E7's chunk-level resumability has no notion of which detectors populated a chunk, so a second detector set pointed at the same output directory would silently skip every image rather than score it. All four detectors' slopes are bootstrapped 2,000 times over images from the same merged records.

On EfficientNet-B6, the apparent asymmetry depends on the control construction: it disappears when the control is additionally matched to the watermark's per-image power spectrum. The watermark–control gap is +0.126 and +0.137 against the PSNR-matched control, both excluding zero; against the spectrum-matched control the same gaps are +0.006 $[-0.009, +0.021]$ and +0.014 $[-0.000, +0.027]$, neither excluding zero. The arms are unchanged, the detector is unchanged, the estimator is

TABLE VI
Evidence-transfer slopes against each control, $n = 800$, 2,000 bootstrap resamples over images. wm−null is the original comparison; wm−spec is the same comparison against the spectrum-matched control. EffNet-B6 is fine-tuned on an undisclosed deepfake corpus; the other three are frozen-backbone linear probes fitted only on this corpus's train split (§IV). Bold marks intervals excluding zero.

Detector	Sch.	wm−null [95% CI]	wm−spec [95% CI]
EffNet-B6	DWT	+.**126** [+.105, +.146]	+.006 [−.009, +.021]
EffNet-B6	Riva	+.**137** [+.116, +.159]	+.014 [−.000, +.027]
ResNet-18	DWT	−.005 [−.023, +.014]	+.012 [−.004, +.028]
ResNet-18	Riva	−.011 [−.031, +.009]	+.**029** [+.013, +.046]
ConvNeXt-B	DWT	+.**037** [+.016, +.057]	+.**039** [+.025, +.054]
ConvNeXt-B	Riva	−.**044** [−.066, −.022]	+.**033** [+.017, +.052]
ViT-B/16	DWT	−.**153** [−.194, −.117]	−.**027** [−.042, −.013]
ViT-B/16	Riva	−.**239** [−.281, −.199]	−.013 [−.030, +.005]

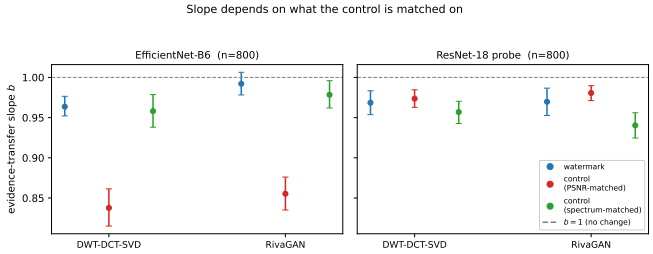


Fig. 3. Slope of each arm with bootstrap intervals. On EfficientNet-B6 the PSNR-matched control sits far below $b = 1$ while the watermark and the spectrum-matched control sit together near it.

unchanged. Only the control's spectrum changed, and the result changed with it.

That closure does not generalise to the other three detectors, and it is not simply absent elsewhere. It reappears in different forms. The ResNet-18 probe is mixed: one contrast excludes zero (RivaGAN, wm−spec +0.029), small and opposite in sign to the EfficientNet-B6 effect; the other does not. ConvNeXt-Base keeps a small, same-signed, statistically reliable wm−spec gap on both schemes (+0.039, +0.033). Spectrum matching narrows the gap from the PSNR-only comparison but does not close it, and the two schemes agree with each other where EfficientNet-B6's controls did not need to. ViT-B/16 differs in kind, not just degree: its wm−null sign is reversed from EfficientNet-B6 (−0.153, −0.239, the watermark attenuating evidence more than the PSNR-matched null, not less), and while spectrum matching shrinks both gaps by roughly an order of magnitude, it still excludes zero for DWT-DCT-SVD. No single story, "the control was inadequate" or "the effect is real", covers all four rows.

The controls are matched in fact, for every detector, because the arms themselves do not change across detectors: only which model scores them changes. Realised PSNR is 39.84 (watermark), 39.84 (PSNR-matched) and 39.83 (spectrum-matched) for DWT-DCT-SVD, and 40.53/40.52/40.45 for RivaGAN. SSIM is closer for the new control than the old one (0.978 vs. 0.962, against a watermark at 0.986), so the stronger control is not winning by being a gentler perturbation, and this holds identically

TABLE VII
Slope against band-limited payload-free perturbation at matched PSNR, EfficientNet-B6, $n = 400$, 2,000 bootstrap resamples.

Arm	slope b	95% CI
watermark	0.9733	[0.9576, 0.9884]
low band	0.9075	[0.8781, 0.9358]
mid band	0.8683	[0.8373, 0.9012]
high band	**0.8092**	[0.7735, 0.8438]

TABLE VIII
Resolution ablation on the ResNet-18 probe, n_{train}=1200, n_{eval}=600, 1,000 bootstrap resamples. EfficientNet-B6 at 528 shown for reference.

Resolution	baseline AUC	wm b	null b	null − wm
224	0.8274	0.9767	0.9828	+0.0061
320	0.8622	0.9676	0.9764	+0.0088
448	0.8767	0.9530	0.9580	+0.0049
528 (EffNet-B6)	0.8703	0.9636	**0.8377**	**−0.1259**

whichever detector's column is read.

We therefore withdraw the reading that imperceptible noise attenuates detector evidence while watermarking does not, as a general claim about passive detection. On EfficientNet-B6 specifically, the defensible statement is narrower and concerns frequency: attenuation grows with the perturbation's frequency band at fixed PSNR (§V-F), neither tested watermark produces much high-frequency energy, and a payload-free perturbation matched to the watermark's full spectrum is indistinguishable from the watermark on that checkpoint. On the other three, a real but smaller and differently-signed gap persists against even the strongest control we built, and we do not have an experiment that explains why (§VII).

F. A frequency dose-response, and what it does not explain

§V-E shows the apparent effect depends on the control's spectrum, but two spectral conditions do not establish a frequency response: the same data are consistent with sensitivity to some property merely correlated with our high-pass noise. We therefore measured the dose-response directly, with payload-free perturbations confined to a single radial band, each binary-searched to the watermark's per-image PSNR (39.77 against the watermark's 39.80), on 400 further held-out images.

The dose-response is monotone in frequency. At fixed PSNR and zero payload, attenuation increases as the perturbation moves up in frequency ($0.908 \to 0.868 \to 0.809$), with the low and high intervals disjoint. This is a measured response rather than an inference from the two conditions of §V-E.

It does not, however, fully account for the watermark. The watermark attenuates less than even the low-band control, and the intervals do not overlap (0.9733 $[0.958, 0.988]$ against 0.9075 $[0.878, 0.936]$). A watermark that is 82% low-band by power is therefore not simply behaving as a low-band perturbation; coarse band membership is too crude a description of what matters.

What does track the watermark is the fidelity of the spectral match. Holding PSNR and payload constant, the control's slope rises monotonically as it matches more of the watermark's spectral structure:

$$\text{PSNR-only } 0.838 \to \text{band-limited low } 0.908 \to \text{full spectrum-matched } 0.958 \to \text{watermark } 0.964$$

(the first, third and fourth from §V-E at $n = 800$; the second from Table VII at $n = 400$, so the ladder mixes two samples and is indicative rather than a single fitted trend). The apparent watermark effect shrinks as the control is matched on more of the perturbation's structure and is statistically indistinguishable from zero once the full per-image spectrum is matched. We stop there: these experiments identify spectral structure as the dimension along which the earlier comparison was mismatched, and do not establish that spectrum is the complete account of this checkpoint's behaviour.

G. Mechanism: resolution is tested and excluded

Why does one checkpoint show this frequency-dependent attenuation and the other not? The candidate we could isolate is input resolution. Corpus images are 256×256; EfficientNet-B6 processes at 528, upsampling and preserving fine perturbation, while the probe processes at 224, downsampling and averaging it away. That predicts the watermark–control contrast should swing negative as resolution rises.

We tested it with everything else held fixed: same backbone weights, same training data, same head, same split, varying only the resolution at which features are extracted, for both the probe fit and the scoring.

Resolution does not produce the asymmetry. Doubling resolution leaves the contrast flat at $+0.005$ to $+0.009$, nowhere near the -0.126 observed at 528 on EfficientNet-B6. What resolution does do is lower both slopes together ($0.977 \to 0.953$ for the watermark arm, $0.983 \to 0.958$ for the control): higher input resolution makes the probe more perturbation-sensitive overall, but symmetrically. Overall sensitivity and the watermark–control asymmetry are separable, and only the second is what an interference study measures.

Resolution is therefore excluded. Architecture, training data and training augmentation remain candidates, and remain untested here: we have one purpose-trained detector whose training pipeline we do not control, so isolating those factors needs a detector panel we could not assemble (§V-A).

H. Payload survival under localised manipulation

The source framework routes "is the watermark present, and has it been altered?" into detection. We supplied the manipulation step that question presupposes and asked where the payload actually fails. One mechanism cannot separate "robust watermarks survive localised editing"

TABLE IX
Percentage of images whose payload remains readable (BER ≤ 0.25) by family and coverage, $n = 200$ per cell. Mean BER in parentheses for the endpoints.

Scheme	Family	5%	10%	25%	50%	75%	100%
DWT	splice	100	100	100	100	97	**17**
DWT	inpaint	100	100	100	100	98	72
DWT	copy-move	100	100	100	98	100	100
DWT	local regen	100	100	100	100	88	36
Riva	splice	100	100	100	100	96	70
Riva	inpaint	100	100	100	100	98	95
Riva	copy-move	100	100	100	100	100	100
Riva	local regen	100	100	100	100	100	97

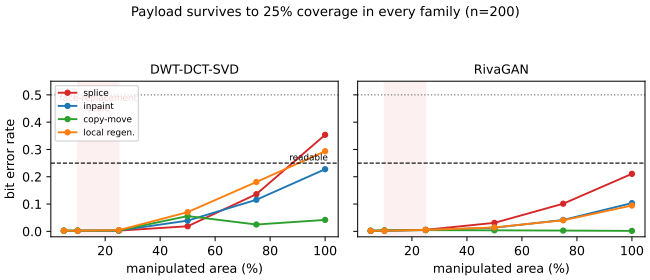


Fig. 4. Mean BER against manipulated area for four manipulation families. The shaded band is realistic face-replacement coverage; the dashed line is the readability threshold.

TABLE X
Expected calibration error by arm and calibrator.

Arm	Platt ECE	Isotonic ECE	slope vs. clean
clean	0.1273	**0.0094**	1.000
DWT-DCT-SVD	0.1228	0.0154	0.968
RivaGAN	0.1353	0.0290	0.990
null[DWT]	0.0958	**0.0492**	0.834
null[RivaGAN]	0.1009	**0.0456**	0.850

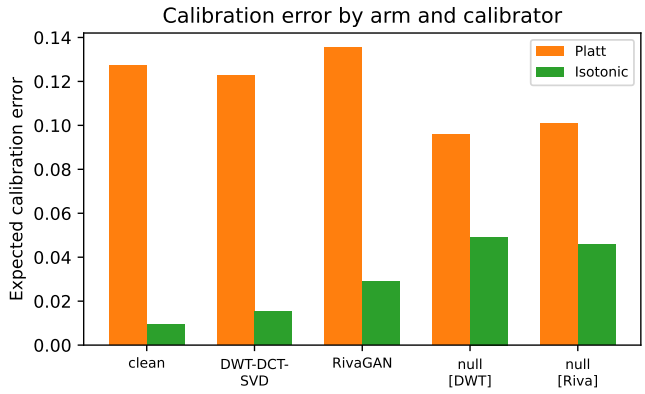


Fig. 5. ECE by arm under each calibrator. Isotonic shows attenuation degrading calibration; Platt shows the opposite, resolved by its badly misspecified clean-arm baseline (0.1273, 13× isotonic's).

from "this splice happens not to disturb the carrier", so we test four families on deliberately different axes: splice (foreign content Poisson-blended in), inpaint (region reconstructed from surrounding context), copy-move (the image's own content relocated over the region), and local regeneration (fine structure destroyed in place by blur and requantisation). Each is swept over a 5–100% coverage grid for both schemes, on 200 held-out images: 9,600 measurements. No detector is involved; BER is measured by extraction alone.

What was measured. Across all eight family–scheme combinations the payload is readable in 100% of images through 25% coverage, the range a face replacement occupies, with mean BER ≤ 0.006. Degradation appears only at $\geq 50\%$ coverage and differs sharply by mechanism: at total coverage DWT-DCT-SVD retains readability in 17% of images under splice but 100% under copy-move, because relocating watermarked content over watermarked content leaves a spatially redundant payload recoverable.

What it suggests. Survival at face-replacement scale is a property of coverage rather than of one manipulation mechanism, and the pattern is consistent with the spatial redundancy expected of a robust scheme. The mechanism matters only once coverage is far past anything a face replacement requires.

What is not established. Two schemes and four hand-constructed families are not the class of robust watermarking, nor the class of manipulations. Our local regeneration is a blur-and-requantise stand-in, not a learned generative edit, and none of these reproduce a modern face swap's detector-facing realism. The consequence we do draw is local to the tested configuration: the provenance feature is effectively constant over the coverage range of interest, so a fused rule over it combines one informative signal with one that does not vary.

I. Calibration under attenuation

Attenuation acts on the score-to-posterior mapping; whether it degrades measured calibration is a separate question, which we test with ECE under Platt and isotonic calibration [20], fit on the clean half of a calibration split and frozen before evaluation (Table X, Fig. 5). Two disclosures about its status. This analysis was not part of pre-registration #2, which committed H1–H3 only; its protocol was fixed before it was run, but it is a secondary analysis, not a pre-registered one. And it reuses the same 20,000 validation images the confirmatory test used, so it is not independent of them. We therefore report it as supporting evidence about the mechanism, not as a second confirmatory result.

The two calibrators disagree, and we report the disagreement rather than selecting the one that suits the argument. Under isotonic regression, attenuation degrades ECE 4–5× relative to clean while watermarking degrades it far less. Under Platt scaling the null arms appear to improve. The disagreement resolves on evidence independent of the comparison: Platt's ECE on the clean arm is already 13× isotonic's (raw logits span −11.6 to +36.8, sd = 7.04), so a single logistic is badly misspecified for these scores before any perturbation is applied, and shrinking

TABLE XI
Decision risk over the cost grid (63 feasible cells): DRD relative to the clean arm. "worse" means the arm's DRD exceeds clean's in that cell.

Calib.	Arm	med. ΔDRD	worse in
isotonic	null[DWT]	+0.0148	**98**%
isotonic	null[Riva]	+0.0042	83%
isotonic	RivaGAN (wm)	**+0.0610**	**100**%
isotonic	DWT-DCT-SVD (wm)	+0.0038	70%
Platt	null[DWT]	−0.0175	8%
Platt	null[Riva]	−0.0151	11%

them happens to partially cancel that misspecification. A ΔECE measured against a broken baseline measures the baseline.

At a single assumed cost triple the decision-risk picture looked like a null: isotonic DRD was 0.0055 (clean) against 0.0145/0.0054 (nulls), small and inconsistent in direction. That reading does not survive a sweep. Evaluating one cost configuration cannot distinguish a robust null from a lucky cell, so we recomputed Chow's thresholds and DRD over an 8×8 grid of (c_{FN}, c_{FP}) at c_R=1; 63 of 64 cells are feasible, the remaining one having an empty review band by construction.

Under isotonic the null arms exceed clean decision risk in 83–98% of the feasible cost space, so the apparent decision-level null was an artefact of the single triple. The largest effect is not a control at all but the RivaGAN watermark arm, worse in every cell with median +0.061. That is consistent with its +0.649 intercept: a translation moves a thresholded decision even where an attenuation does not, which is precisely the asymmetry $\Delta\mu$ and slope were introduced to separate. Under Platt the pattern does not hold, as expected from a calibrator whose clean-arm baseline is already broken.

We state this carefully. It is a decision-level effect associated with one watermark's score translation, measured post hoc on data not independent of the confirmatory test, over an assumed rather than elicited cost space. It is not evidence of watermark-specific interference in the sense §V-E tests and rejects: the slope comparison against a spectrum-matched control is the test of that, and it is null.

J. The fusion comparison is not estimable; an exploratory transform analysis

Two distinct points, which we separate deliberately because conflating them would misrepresent the second.

First, the planned fusion comparison (F0–F5, with and without the interaction term of Eq. 4) under localised manipulation is not estimable in this setup. §V-H shows the provenance feature is effectively constant across the manipulation coverage of interest, and a coefficient on a feature that does not vary is not a weak estimate but an undefined one. We therefore report no confirmatory fusion result, and nothing below is offered as a substitute for it.

Second, and separately, we ran an exploratory analysis asking a narrower question: does the interaction term respond to benign input changes at all, on otherwise unmanipulated images? This is reported solely to characterise the term's behaviour, not to establish any fusion claim. Parameters: n=400 (120 calibration / 280 test, held-out test split, disjoint from validation), EfficientNet-B6, seven transforms, and no bootstrap intervals. Point estimates only, at a sample size an order of magnitude below the confirmatory analyses. Results are in Table XII.

TABLE XII
Exploratory transform analysis, F0 (no interaction) vs. F4 (with it). Metrics are computed on the 280 held-out test images of a 400-image stratified split (120 calibration / 280 test). Point estimates only: no bootstrap intervals were computed, and no cell here is claimed as a result.

Transform	F0 AUC	F0 DRD	F4 AUC	F4 DRD	F4 β_4
JPEG(90)	0.899	0.006	0.897	0.008	+0.004
JPEG(70)	0.776	0.003	0.819	0.005	−0.022
JPEG(50)	0.721	0.001	0.794	0.005	+0.038
Resize(0.75)	0.743	0.003	0.825	0.008	+0.048
Resize(0.5)	0.710	0.001	0.825	**0.050**	**+0.221**
Bright(+20%)	0.909	0.034	0.909	0.031	+0.056
Bright(−20%)	0.841	0.003	0.848	0.006	−0.072

F0 and F4 AUC fall together under compression and downscaling, which tracks the detector's own fragility rather than anything watermark-specific. β_4 stays small ($|\beta_4| < 0.08$) for six of the seven transforms. Resize-to-0.5× is the exception ($\beta_4 = +0.221$, DRD 0.050). With n=280 test images and no interval, we flag that cell as the one worth a CI-backed rerun and claim nothing from it. A ρ sweep (fusion fit once on calibration, then evaluated on mixtures at watermarked fractions $\rho \in \{0.05, 0.25, 0.5, 1.0\}$) found AUC, ECE and DRD essentially flat across ρ for both schemes, which is what the rest of the paper would predict: a feature that does not move the outcome at ρ=1 will not start mattering because it appears less often. Full per-ρ values are in the released results.

VI. Limitations

We distinguish limitations we removed by experiment from those that remain, because the two have different consequences for a reader.

Removed: the control matched only energy. This was the deepest limitation of the earlier design and it is now addressed by construction (§V-D–V-E) rather than by caveat. The residual limitation is narrower: spectrum matching is second-order, so a watermark could still differ from the control in higher-order or phase structure that our arm randomises away. What we can say is that the apparent effect does not survive matching on energy and spectrum.

Removed: one assumed cost triple. Replaced by a 63-cell sweep (§V-I), which reversed the conclusion drawn from the single cell. The costs are still assumed rather than elicited from a deployment; what the sweep buys is that the conclusion no longer depends on which cell we happened to choose.

Reduced: the mechanism was unresolved. Input resolution is now tested and excluded (§V-G). Architecture, training data and training augmentation remain untested, so the mechanism is narrowed to "frequency-dependent attenuation specific to this checkpoint" rather than identified. §V-F also shows frequency band alone does not account for the watermark's own position, so even the frequency account is partial.

Reduced: a two-detector panel. §V-E widens the panel to four detectors across three architecture families, which is enough to show the phenomenon is architecture-dependent rather than universal or absent. It is not enough to say which population of detectors exhibits which pattern, or why: three architecture families is a small sample of architecture space, all four detectors share one corpus and two watermarking schemes, and §V-A shows floor-clearing public detectors are scarce (five of six screened checkpoints operate at chance on our distribution), so the three added probes are self-trained rather than drawn from a wider public population. New: the mechanism behind the fracture is unresolved. §VII names two untested hypotheses (kernel-size sensitivity to local phase structure; attention-level payload disruption) and we have run neither the kernel ablation nor the attention analysis that would distinguish them, so the architecture-dependence itself joins the list of narrowed-but-not-identified mechanisms alongside §V-F's frequency account.

Remains: the strong detector's provenance. EfficientNet-B6's training data is undisclosed, so contamination with our evaluation distribution cannot be excluded. We mitigate with within-model, within-image deltas, under which common-mode memorisation cancels to first order. Resolving it requires information only the checkpoint's authors hold.

Remains: the β_4 outlier lacks an interval. The resize-to-0.5× cell of the exploratory transform analysis (§V-J) is a point estimate at n=280. We did not rerun it with bootstrap intervals, so it stays flagged and unclaimed.

Remains: synthesis type and dependence structure. The corpus is whole-image GAN synthesis against real photographs, not face-swap or reenactment manipulation [21], and it carries no identity metadata, so intervals are image-level bootstraps and no identity-clustered resampling is possible. Neither is fixable by analysis.

VII. Discussion

The control is the experiment. The substantive finding here is not a number about watermarking. It is a fact about how this class of experiment is built. Across §V-E every arm, detector and estimator was held fixed. The only thing that changed was what the control was matched on, and the conclusion inverted. A PSNR-matched control is the obvious choice. It is what we chose, and it is not sufficient, because fixing energy leaves the perturbation free to occupy the opposite end of the spectrum. In our case it did.

Architecture, not just control quality, determines what a fixed comparison finds. §V-E's four-detector panel forecloses the reading that "an adequate control makes the effect go away" as a general statement: it does on EfficientNet-B6, does not on ConvNeXt-Base, and produces a sign-reversed effect on ViT-B/16 that spectrum matching only partially closes. We can describe the pattern without explaining it. Two hypotheses are consistent with the pattern and neither is tested here, so we name them as open questions rather than findings. First, ConvNeXt's large-kernel depthwise convolutions could be sensitive to localised spatial phase structure that a radially-averaged spectral match does not constrain. Our spectrum-matched control fixes the magnitude of the 2-D Fourier transform and randomises phase globally, which need not preserve whatever local phase coherence a depthwise kernel responds to. Second, ViT-B/16's patch tokenisation and global self-attention process an image differently in kind from any convolution, and the sign reversal, the watermark attenuating evidence more than an energy- and spectrum-matched control rather than less, is consistent with the payload perturbing patch-level or attention-level structure that pure noise does not. But we have no attention-map analysis, no patch-level ablation, and no experiment that isolates tokenisation or attention as the cause rather than correlates with it. Both explanations also compete with the confound noted in §IV: EfficientNet-B6 is the only fine-tuned, task-specific detector in the panel, so "architecture" and "training regime" are not independently varied between it and the three probes, only among the three probes themselves. Resolving any of this needs an experiment that holds architecture fixed and varies training regime, or vice versa, and holds a kernel-size or attention ablation fixed against the same controls. We have run neither and report the fracture as a result in search of a mechanism, not a mechanism.

What a null does and does not mean. We do not detect [1]'s effect once the control is adequate, which is a boundary condition rather than a refutation. Their detectors are trained for the manipulation family they evaluate and their images are manipulated after marking, so watermark and forgery signals coexist in the way their overlap argument describes. Ours are zero-shot detectors on whole-image synthesis marked after the fact. What our result does imply for that literature is procedural: a reported interference effect measured against an energy-matched control cannot rule out spectral confounding unless the relevant residual spectra are examined.

Our own gate had the asymmetry it warns about. We gated on $\Delta\mathrm{AUC}_{net}$ while the quantity a calibration-centred thesis needs is $\Delta\mu_{net}$; every reading of the latter is post-hoc with respect to that gate, which is why it was re-registered before being tested rather than reinterpreted in place. The same discipline is why §V-I's cost sweep is labelled secondary: it reuses data the confirmatory test already used.

One checkpoint is not a detector, and one cell is not

a cost model. Three of this paper's corrections came from widening a design that had collapsed to a single point. One checkpoint became two, and a generality claim reversed. Two became four, and a second claim, that an adequate control closes the gap, narrowed from general to checkpoint-specific in the same widening. One cost triple became sixty-three cells, and a decision-risk null reversed. None of the three widenings was expensive. The pattern is worth more than any single result: every generality claim in this paper that was checked against a wider design either reversed or narrowed, and none survived widening unchanged.

VIII. Conclusion

We set out to measure whether provenance watermarking interferes with passive synthetic-image detection. Against a payload-free control matched per image to the watermark's PSNR, the answer looked clear: on 20,000 held-out images an EfficientNet-B6 detector's evidence is attenuated $\approx 16\%$ under the control and left near identity under watermarking, so the degradation is not attributable to the watermark.

That conclusion does not survive a better control, on this checkpoint. The PSNR-matched arm is spectrally inverted relative to the watermark it stands in for: both tested watermarks place 0.83–0.97 of their residual power in the lowest radial band, and their controls place 0.46 in the highest. A payload-free arm carrying the watermark's own spectrum closes the gap from $+0.126$ $[+0.105, +0.146]$ to $+0.006$ $[-0.009, +0.021]$ on EfficientNet-B6. With the arms, detector and estimator unchanged, matching the control's spectrum removes the detectable watermark–control slope difference on that checkpoint. The remaining gap is not distinguishable from zero there.

It does not close the same way everywhere. Scoring the identical 800-image slice through two more disclosed-provenance probes, ConvNeXt-Base and ViT-B/16, frozen backbones with linear heads trained the same way as the ResNet-18 probe, shows the closure is architecture-dependent. ConvNeXt-Base keeps a small, statistically reliable, same-signed gap against the spectrum-matched control on both watermarking schemes ($+0.039$, $+0.033$, both excluding zero). ViT-B/16 reverses which arm attenuates more even before spectrum matching (-0.153, -0.239 against the PSNR-matched null, the opposite sign from EfficientNet-B6), and spectrum matching narrows but does not fully close that gap either. We name two untested hypotheses for the fracture: kernel-size sensitivity to local phase structure the spectrum-matched control does not constrain, and payload disruption of attention-level structure a frozen linear head cannot itself explain. We report neither as established, because EfficientNet-B6 is fine-tuned on an undisclosed corpus while the other three are frozen probes on ours, so architecture and training regime are not independently varied across the full panel. The corrected claim is not "the effect disappears with an adequate control." It is "whether the effect survives an adequate control depends on the detector, and we can report the pattern without yet explaining it."

On EfficientNet-B6 specifically, the residual phenomenon is narrower and belongs to a model rather than to detection generally: attenuation grows monotonically with the perturbation's frequency band at fixed PSNR (0.908/0.868/0.809 for low/mid/high), the ResNet-18 probe matched to within 0.003 baseline AUC does not show it in a full-scale replication, and a resolution sweep from 224 to 448 shows that input resolution does not explain it over that range on that pair, while leaving architecture, training data and augmentation untested. Three supporting measurements accompany this: both tested watermarks keep a readable payload through 25% manipulated area across four manipulation families (9,600 measurements), so the provenance feature is constant over the range a face replacement occupies; attenuation degrades isotonic calibration, and over a 63-cell cost grid the decision-risk null we previously reported does not hold; and five of six screened public checkpoints operate at chance on our distribution.

Two results are analytic and independent of all of this: a mean-shift statistic cannot distinguish attenuation from translation, and a strictly increasing rescaling leaves AUC exactly unchanged, so a ranking-based gate cannot register attenuation at all.

None of this shows that watermark–detector interference is absent in general, and none of it shows that robust watermarks cannot evidence tampering. What it shows is that an interference study's conclusion can be determined by a property of its control that the matching criterion never constrained, and that the standard criterion does not constrain it.

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